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-- PART 4: RIDES TABLE WITH YEAR-BASED PARTITIONING
CREATE TABLE IF NOT EXISTS rides (
  ride_id BIGINT NOT NULL AUTO_INCREMENT,
  external_id CHAR(36) NOT NULL DEFAULT (UUID()),
  user_id BIGINT NOT NULL,
  driver_id BIGINT DEFAULT NULL,
  vehicle_id BIGINT DEFAULT NULL,
  city_id INT NOT NULL,
  pickup_zone_id INT DEFAULT NULL,
  dropoff_zone_id INT DEFAULT NULL,
  status
ENUM('requested','accepted','driver_en_route','in_progress','completed','cancelled','disputed') NOT
NULL DEFAULT 'requested',
  pickup_lat DECIMAL(10,8) NOT NULL,
  pickup_lon DECIMAL(11,8) NOT NULL,
  pickup_address_encrypted BLOB DEFAULT NULL,
  dropoff_lat DECIMAL(10,8) NOT NULL,
  dropoff_lon DECIMAL(11,8) NOT NULL,
  dropoff_address_encrypted BLOB DEFAULT NULL,
  estimated_distance_km DECIMAL(8,3) DEFAULT NULL,
  actual_distance_km DECIMAL(8,3) DEFAULT NULL,
  estimated_duration_minutes SMALLINT DEFAULT NULL,
  actual_duration_minutes SMALLINT DEFAULT NULL,
  base_fare DECIMAL(10,2) DEFAULT NULL,
  surge_multiplier DECIMAL(4,2) NOT NULL DEFAULT 1.00,
  distance_fare DECIMAL(10,2) DEFAULT NULL,
  time_fare DECIMAL(10,2) DEFAULT NULL,
  total_fare DECIMAL(10,2) DEFAULT NULL,
  driver_earnings DECIMAL(10,2) DEFAULT NULL,
  platform_commission DECIMAL(10,2) DEFAULT NULL,
  payment_method ENUM('cash','telebirr','cbe_birr','awash_pay','amole','card') NOT NULL DEFAULT
'cash',
  requested_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  accepted_at DATETIME DEFAULT NULL,
  driver_arrived_at DATETIME DEFAULT NULL,
  started_at DATETIME DEFAULT NULL,
  completed_at DATETIME DEFAULT NULL,
  cancelled_at DATETIME DEFAULT NULL,
  cancellation_reason VARCHAR(255) DEFAULT NULL,
  cancelled_by VARCHAR(50) DEFAULT NULL,
  user_rating TINYINT DEFAULT NULL,
  driver_rating TINYINT DEFAULT NULL,
  user_feedback TEXT DEFAULT NULL,
  is_scheduled TINYINT(1) NOT NULL DEFAULT 0,
  scheduled_for DATETIME DEFAULT NULL,
  version INT NOT NULL DEFAULT 1,
  ride_year SMALLINT NOT NULL,
  PRIMARY KEY (ride_id, ride_year),
  UNIQUE KEY uq_ride_external (external_id, ride_year),
  KEY idx_ride_user (user_id, requested_at),
  KEY idx_ride_driver (driver_id, requested_at),
  KEY idx_ride_city (city_id, requested_at),
  KEY idx_ride_status (status),
  KEY idx_ride_pickup (pickup_lat, pickup_lon),
  KEY idx_ride_completed (completed_at),
  CONSTRAINT chk_user_rating CHECK (user_rating BETWEEN 1 AND 5),
  CONSTRAINT chk_driver_rating_val CHECK (driver_rating BETWEEN 1 AND 5),

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    CONSTRAINT chk_surge CHECK (surge_multiplier >= 1.00 AND surge_multiplier <= 10.00)
) ENGINE=InnoDB
PARTITION BY RANGE (ride_year) (
    PARTITION p2023 VALUES LESS THAN (2024),
    PARTITION p2024 VALUES LESS THAN (2025),
    PARTITION p2025 VALUES LESS THAN (2026),
    PARTITION p2026 VALUES LESS THAN (2027),
    PARTITION p_future VALUES LESS THAN MAXVALUE
);
```